

Introduction to NLP: Text, Tokens & Representations

KAUST–Mawhiba IOAI Training

Week 3 · Day 12



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1. From Vision to Language
2. What is NLP?
3. NLP Pipeline and Core Tasks
4. Text Preprocessing
5. Corpora and Vocabulary
6. Tokenization and N-grams
7. Bag of Words
8. Summary

- ▶ **Week 2** took us deep into Computer Vision:
 - CNNs, pretrained models (ResNet, MobileNet, EfficientNet)
 - Transfer learning & fine-tuning
 - Autoencoders, YOLO, U-Net
- ▶ All of these deal with **images**: grids of pixel values
- ▶ But what about **text**?
 - Reviews, articles, tweets, code, conversations...
 - Text is **sequential**, **symbolic**, and **variable-length**

1. What is Natural Language Processing (NLP)?
2. The NLP pipeline and core tasks
3. Text preprocessing: cleaning raw text
4. **Tokenization**: breaking text into units
5. **Vocabulary** and corpora
6. N-grams: capturing local context
7. Bag of Words: turning text into numbers

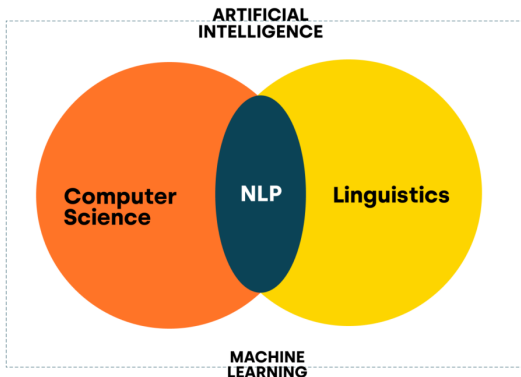
Goal: Understand how to go from raw text → numerical features a model can learn from.

Natural Language Processing: **Introduction**

What is NLP?

- ▶ Study of computational approaches to processing natural languages.
- ▶ Processing includes:
 - Acquiring language data
 - Representing information
 - Storing text and speech
 - Understanding meaning
 - Characterizing language patterns
 - Generating new language
- ▶ Natural languages refer to human languages.

What is Natural Language Processing?



NLP = Computer Science + Linguistics + AI

► Deals with:

- Language understanding (input)
- Language generation (output)
- Language translation
- Information extraction

► Subfields:

- Syntax
- Semantics
- Pragmatics
- Discourse



Goal: Deep Understanding

Requires context, linguistic structure, meanings...



To Avoid: Shallow Matching

Could be useful also though depending on use case

Goal of NLP:

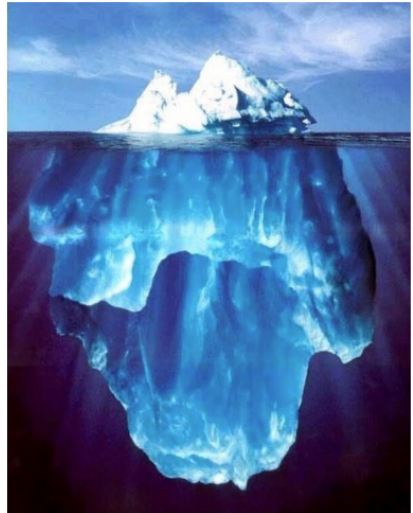
- ▶ Enable machines to understand and generate human language.
- ▶ Facilitate human-computer interaction through natural language.
- ▶ Develop systems that can process and analyze large amounts of text data.

- ▶ **Text Preprocessing**
 - Cleaning and preparing raw text for analysis.
- ▶ **Tokenization**
 - Splitting text into words, sentences, or other meaningful units.
- ▶ **POS Tagging**
 - Assigning parts of speech (noun, verb, etc.) to each token.
- ▶ **Parsing**
 - Analyzing grammatical structure of sentences.
- ▶ **Named Entity Recognition (NER)**
 - Identifying entities such as people, organizations, locations.
- ▶ **Sentiment Analysis / Classification**
 - Determining sentiment or categorizing text.
- ▶ **Language Modeling**
 - Predicting the next word or sequence in text.

- ▶ An iceberg is a large piece of freshwater ice that has broken off from a snow-formed glacier or ice shelf and is floating in open water.



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Teaching machines to **understand and generate human language**

► **Text preprocessing:**

Cleaning and preparing raw text data (removing noise, tokenization, normalization).

► **Understanding meaning:**

Extracting meaning from text using techniques like part-of-speech tagging, named entity recognition, and sentiment analysis.

► **Language modeling:**

Building models that can predict or generate text, such as autocomplete or next-word prediction.

► **Translation, summarization, etc.:**

Enabling applications like machine translation, text summarization, question answering, and more.

Text data is everywhere: tweets, reviews, articles, chats! NLP helps us make sense of this vast information.

Task	What It Does	Example
Tokenization	Split text into words	"I love NLP" → ["I", "love", "NLP"]
POS Tagging	Label grammar tags	"Dogs bark" → [Noun, Verb]
Named Entity Recognition	Find names, places, etc.	"Christopher Nolan lives in Los Angeles"
Sentiment Analysis	Detect mood	"This movie was amazing!" → Positive
Machine Translation	Language to language	English → French

► **Lowercase everything:**

Example: "NLP" → "nlp"

► **Removing stop words:**

Eliminate common words that carry little meaning (e.g., "is", "the", "and") to focus on important content.

► **Stemming/Lemmatization:**

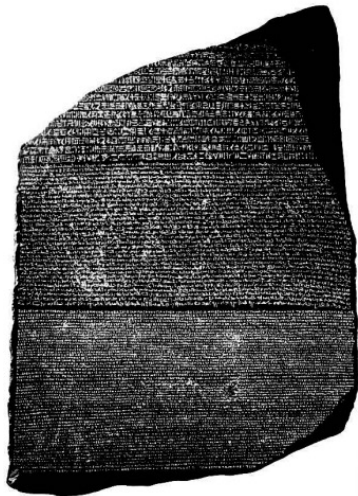
Reduce words to their root or base form.

Example: "running" → "run"

► **Vectorization:**

Transform words or documents into numerical representations for machine learning models:

- **Bag of Words:** Counts word occurrences in a document.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Weighs words by importance across documents.
- **Word2Vec:** Learns dense vector representations capturing word meaning and



What is a corpus?

- ▶ A corpus is a collection of text.
- ▶ Often annotated in some way.
- ▶ Sometimes just lots of text.
- ▶ **Balanced corpora:** Usually not possible in practice.
- ▶ **Examples:**
 - Newswire collections: 500M+ words
 - Brown corpus: 1M words of tagged “balanced” text
 - Penn Treebank: 1M words of parsed WSJ
 - Canadian Hansards: 10M+ words of aligned French/English sentences
 - The Web: billions of words of who knows what

- ▶ **Vocabulary** = All unique words in your dataset
- ▶ Example: "I love NLP and NLP loves me" →
Vocabulary = {"I", "love", "NLP", "and", "loves", "me"}
- ▶ More data = Bigger vocabulary = Harder to process!

💡 Tip: Rare words may not help; common words may not mean much.

Traditional approach: One-hot encoding

- ▶ Example: "NLP" $\rightarrow$ [0, 0, 1, 0, 0, 0, 0...]

Problem:

- ▶ High-dimensional (thousands of words!)
- ▶ Sparse (mostly 0s)
- ▶ No meaning in structure (no relation between "king" and "queen")
- ▶ ✗ Not efficient for learning

Feature extraction = Turning text into numbers

- ▶ Count how often each word appears (**Term Frequency**)

Example:

Text	"great product"	"bad product"
Word: "great"	1	0
Word: "bad"	0	1

 Use this to find patterns in sentiment, spam, etc.

Suppose you're classifying reviews:

Positive Reviews: ["amazing", "good", "great"]

Negative Reviews: ["bad", "awful", "terrible"]

Count how often each word appears in each class.

Example table:

Word	Positive Count	Negative Count
good	20	1
bad	1	30

Helps models detect the “tone” (**sentiment clues**) of new text.

- ▶ **Bag of Words (BoW):** Just counts word frequencies in each document.
- ▶ **TF-IDF (Term Frequency-Inverse Document Frequency):** Adjusts for how “unique” or important a word is in a document compared to all documents.
 - Words like "the", "is" are less important.
- ▶ **Word Embeddings (later):** Add meaning and capture relationships between words (e.g., similarity, analogy).

Natural Language Processing: **N-grams**

What are N-grams?

An N-gram is a sequence of N words

Corpus: I am happy because I am learning

Unigrams: { I , am , happy , because , learning }

Bigrams: { I am , am happy , happy because ... }

✗ I happy

Trigrams: { I am happy , am happy because, ... }

- Items are typically **words** or **characters**.

- ▶ Tokenization is the process of breaking text into smaller units called **tokens**.
- ▶ Tokens can be words, characters, or subwords.
- ▶ Example: "I love NLP" can be tokenized into:
 - Words: ["I", "love", "NLP"]
 - Characters: ["I", " ", "l", "o", "v", "e", " ", "N", "L", "P"]
 - Subwords: ["I", " ", "lov", "e", " ", "N", "L", "P"]
- ▶ Tokenization is crucial for preparing text data for NLP tasks.

Sentence: “I love NLP”

- ▶ **Unigrams:** I, love, NLP
- ▶ **Bigrams:** I love, love NLP
- ▶ **Trigrams:** I love NLP

- ▶ Capture local word co-occurrence
- ▶ Build simple language models
- ▶ Easy to compute and analyze
- ▶ Trade-off between simplicity (unigram) and contextual richness (trigram)

Idea: Represent each document as a vector of word counts.

Ignore word order – just count what appears!

Example:

- ▶ Doc 1: “I love NLP”
- ▶ Doc 2: “I love cats”

	I	love	NLP	cats
Doc 1	1	1	1	0
Doc 2	1	1	0	1

Given corpus:

1. “The cat sat on the mat”
2. “The dog sat on the log”
3. “The cat chased the dog”

Step 1: Build vocabulary = {the, cat, sat, on, mat, dog, log, chased}

Step 2: Count occurrences per document:

	the	cat	sat	on	mat	dog	log	chased
Doc 1	2	1	1	1	1	0	0	0
Doc 2	2	0	1	1	0	1	1	0
Doc 3	2	1	0	0	0	1	0	1

Each row is a **document vector** in $\mathbb{R}^{|V|}$.

Training data:

Review	Label
"great movie loved it"	Positive
"terrible movie hated it"	Negative
"loved the great acting"	Positive

BoW features → classifier (Logistic Regression, SVM, etc.)

The classifier learns that ``great'', ``loved'' correlate with Positive and ``terrible'', ``hated'' correlate with Negative.

- ▶ **✗ Ignores word order:** “dog bites man” = “man bites dog”
- ▶ **✗ High-dimensional:** vocabulary can be 50K+ words
- ▶ **✗ Sparse:** most entries are 0
- ▶ **✗ No semantics:** “happy” and “joyful” are unrelated vectors
- ▶ **✗ Common words dominate:** “the”, “is”, “a” get high counts but carry little meaning

💡 **Solution:** TF-IDF and word embeddings (Day 13!)

- ▶ **NLP** enables machines to understand and generate human language
- ▶ The NLP pipeline: preprocessing → tokenization → feature extraction → modeling
- ▶ **Tokenization** splits text into words, characters, or subwords
- ▶ **Vocabulary** is the set of all unique tokens in a corpus
- ▶ **N-grams** capture local word co-occurrence and context
- ▶ **Bag of Words** turns text into numerical vectors by counting

- ▶ BoW counts words but misses **meaning**
- ▶ **TF-IDF**: smarter weighting – important words get higher scores
- ▶ **Vector Space Models**: representing meaning geometrically
- ▶ **Word Embeddings**: dense vectors that capture semantic relationships
 - Word2Vec, GloVe, fastText
 - “king” – “man” + “woman” $\approx$ “queen”

From counting words → understanding meaning!

Credits

KAUST Academy

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NLP introduction content adapted from
KAUST Academy NLP course materials

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